

QUANTUM MACHINE LEARNING · QML-2026 FDP · SESSION 2

teaching machines to see

classical & quantum-enhanced computer vision
— from convolution to quanvolution

Archit Srivastava · E&ICT Academy NIT Warangal × NIT Raipur · 11 July 2026

ABOUT THE SPEAKER

Archit Srivastava

5+
research papers

25+
invited talks

10+
universities

200+
faculty engaged

- quantum × ai

photonic hardware

gw & black holes

enterprise tech

- Senior Technical Consultant, **o9 Solutions** — enterprise AI & decision systems
- Founder, **AiQyaM** — India's first quantum-hardware community, est. **2020**
- Founder, **CIRQuIT Quantum Research**, RV College of Engineering
- Senior Research Associate, **Quantum Computing India**
- **GKQCTP & IQDC** with Innogress Ventures — toward a sovereign Quantum Tech Park
- Quantum PoCs at **HPE** & **o9** · early intern **BosonQ Psi** · **Conf42** speaker
- Published: **Quantum Finance** (VQE, Qiskit vs PennyLane) & **AI computer vision** with a quantum-visual-tracking outlook

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CURRENT FOCUS

From a grid of numbers to a quantum feature.

1 · What an image is

A spreadsheet of brightness — and why raw pixels are a terrible representation.

2 · The CNN

The convolution atom, and how features learn themselves. Built layer by layer.

3 · The quantum turn

Encoding an image, and replacing the convolution atom with a quantum circuit.

4 · The honest picture

A live demo, the QCNN, where quantum vision is real, and India's stake.

WHY NOW · 2025 WAS THE YEAR QUANTUM GOT REAL

From promise to measured advantage.

Three peer-reviewed firsts in a single year moved quantum advantage from a talking point to a number you can check.

Google Willow

0

- superconducting qubits, **Nature** Dec 2024
- first **below-threshold** error correction — logical error shrinks as the code grows
- benchmark in <5 min vs $\sim 10^{25}$ yr classically

D-Wave Advantage2

$0+$

- qubit annealer, **Science** Mar 2025
- magnetic-materials simulation in minutes
- a **real scientific target** — parts later contested on GPUs

Quantum Echoes

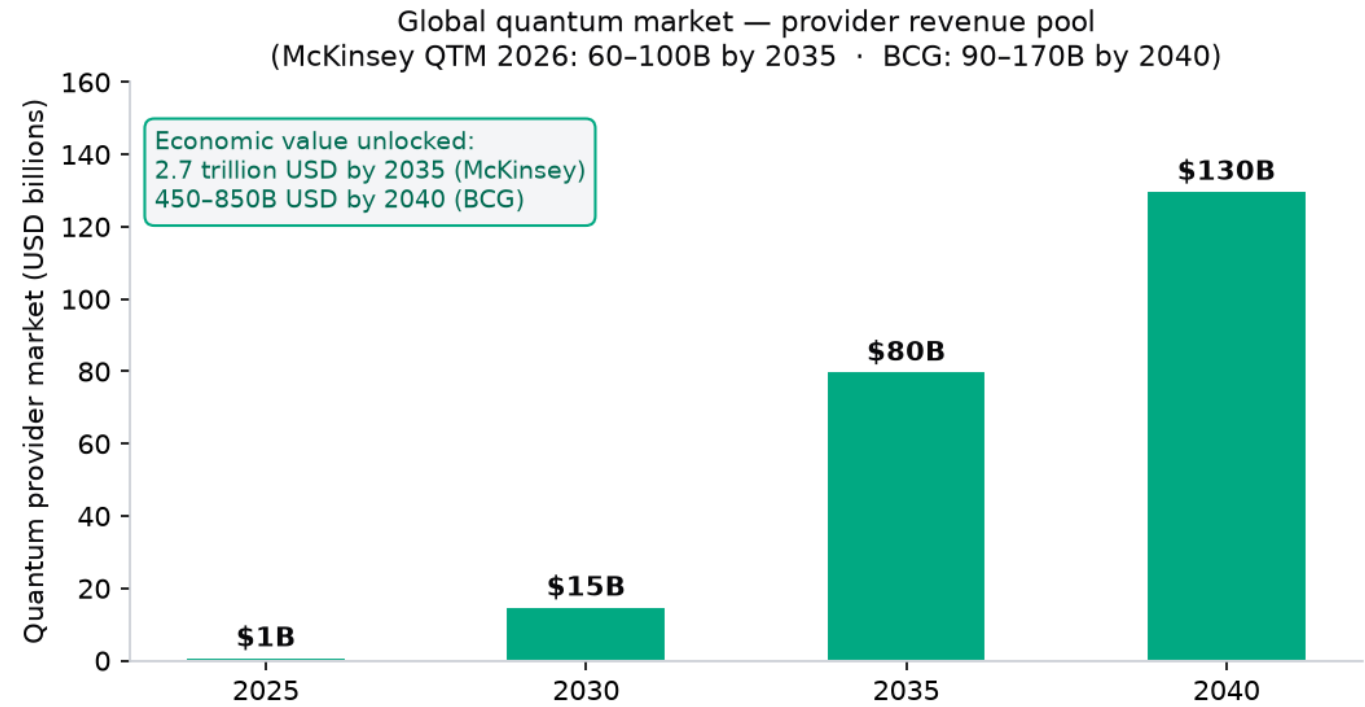
$0\times$

- Google, **Nature** Oct 2025
- first **verifiable** advantage — checkable on a second machine
- a path toward molecular-structure readouts

The shift that matters for this room: the field has crossed from "can a quantum computer do anything hard?" to "on which problems, and can we verify it?" That is the question your students will spend their careers answering.

The value isn't in selling processors — it's in the outcomes.

GLOBAL MARKET · MCKINSEY QTM 2026 · BCG



THE SIGNAL

- **\$2.7 trillion** economic value unlocked by 2035 (McKinsey)
- internal market **\$60–100B by 2035**; QC alone \$43–71B
- start-up investment **\$12.6B in 2025** — **6.3x YoY**, ~90% to quantum computing
- industry needs **~840,000 workers by 2035**; ~5,000 qualified in 2025
- the value is captured by **software, algorithms & AI** — a keyboard-and-brains problem

\$2.7T

economic value by 2035 (McKinsey)

6.3×

YoY jump in quantum investment, 2025

~840k

quantum jobs needed by 2035 — the talent gap

India's National Quantum Mission

A national bet across mission funding, thematic hubs and state valleys — building hardware, secure networks and talent so India isn't dependent on a foreign quantum stack.

The mission

₹6,003 cr

- approved **2023** · 8-year horizon to **2031**
- staged target **50–1,000 physical qubits**
- QKD to **2,000 km**; sensors, atomic clocks, materials
- ~half the targets reportedly met within **3 years**

Four thematic hubs · 43 institutions

- **Computing** — IISc Bengaluru
- **Communication** — IIT Madras + C-DOT
- **Sensing** — IIT Bombay
- **Materials** — IIT Delhi
- 152 researchers · 17 states — a **national footprint**

Shipping, not slideware

0 qubits

- **QpiAI** "Kaveri" 64-qubit (Nov 2025); "Indus" 25-qubit
- **IISER Pune** ion-trap, 20 qubits due end-2026
- **Amaravati Quantum Valley** — IBM System Two, launched Feb 2026
- ~**1,000 km** quantum comms demonstrated · national **PQC roadmap**

FIRST PRINCIPLES · THE RAW MATERIAL

The computer never sees a cat.

It sees a spreadsheet of brightness values. Computer vision is the art of turning that grid back into meaning.

AN IMAGE IS A GRID OF NUMBERS

- Greyscale = a matrix of intensities (0 black ... 255 white)
- Colour = three such matrices (R, G, B)
- A 28x28 digit = **784 numbers**. A 1-MP colour photo = **3 million**.

WHY RAW PIXELS ARE TERRIBLE

- **High-dimensional** — hundreds to millions of features
- **No built-in locality** — neighbouring pixels are just two flat entries
- **Not translation-invariant** — shift the cat 2px and every number changes

Vision is about learning good representations: local, hierarchical, invariant.

Convolution: slide a small window, combine.

THE OPERATION

Slide a small kernel (e.g. 3×3) over the image; at each stop, compute a weighted sum. Different kernels detect different things — a **Sobel** kernel finds edges, a blur kernel averages.

$$(I * K)(i, j) = \sum_m \sum_n I(i+m, j+n) \cdot K(m, n)$$

This "**slide-a-window-and-combine**" operation is the atom of all of vision.

Remember it — the quantum method replaces exactly this atom.

The CNN — features that learn themselves.

AlexNet (2012) halved ImageNet error and started the deep-learning era. The key idea: don't hand-design the kernels — **learn them from data**.

Conv layer

Many learnable kernels slide over the image → **feature maps**. Edges → textures → objects.

ReLU

Keep positive responses, zero the rest → models non-linear patterns.

Pooling

Max/average over a window
→ shrinks the map, buys **translation invariance**.

FC + softmax

Flatten final features → classify. Trained end-to-end by backprop.

Three structural priors make CNNs work: local connectivity (a pixel relates to its neighbours) · weight sharing (one edge-detector, useful everywhere → far fewer parameters) · hierarchy (simple features compose into complex ones). These recur in the quantum story.

State-of-the-art vision is enormous.

Beyond CNNs

Vision Transformers (2020) treat an image as patches and use attention — strong at scale, hungrier still.

The bill

Frontier models cost **millions of dollars** and **megawatt-hours** to train, on oceans of labelled data.

The opening

Quantum-enhanced methods aim here: **richer features from fewer parameters**, or feature spaces classical nets can't cheaply reach.

How do you feed an image to a quantum computer?

Data encoding is 80% of quantum ML for vision. Three schemes — a trade-off between qubits and cost.

Basis

Classical bitstring → computational basis state.

N bits → N qubits

Wasteful; rarely used.

Angle ★

Each feature → a rotation angle $R_y(x_i)$ on a qubit.

N features → N qubits

Shallow, NISQ-friendly. How real images reach real hardware today.

Amplitude

Features → amplitudes of the state vector.

N features → $\lceil \log_2 N \rceil$ qubits

Exponential compression — but expensive to prepare (QRAM).

Angle encoding turns one small patch of pixels into a few rotation angles — that's what makes quanvolution run today. **Amplitude encoding** is the exponential dream, waiting on hardware.

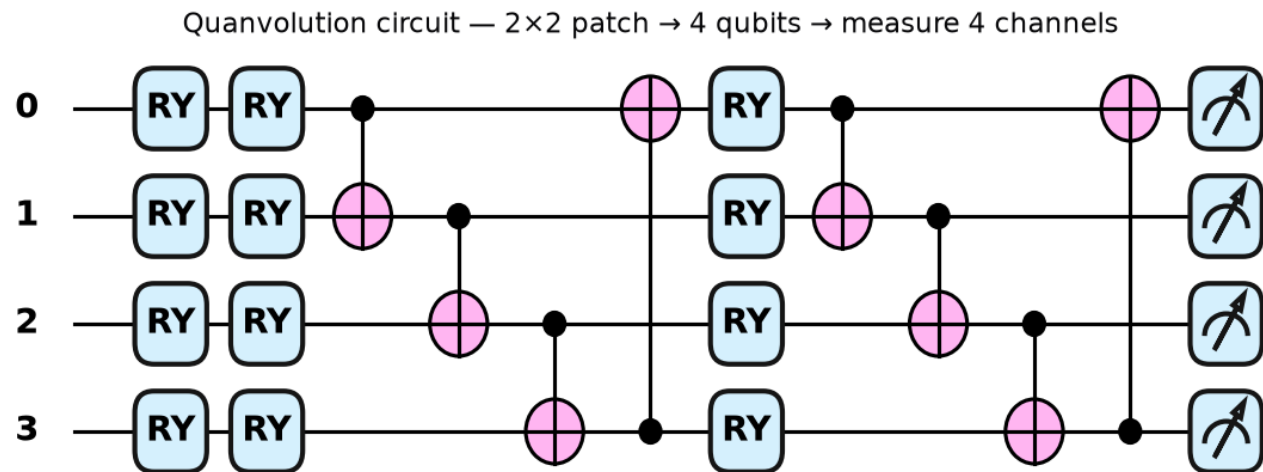
Quanvolution — replace the atom with a circuit.

THE RECIPE (HENDERSON ET AL., 2019)

- Slide a **2×2 window** — exactly like classical convolution
- **Angle-encode** the 4 pixels on 4 qubits
- Apply a **random/trainable circuit** — entangling gates mix pixels in a 16-D Hilbert space
- **Measure** each qubit → new feature-map channels
- Stack them, feed a **classical CNN head**

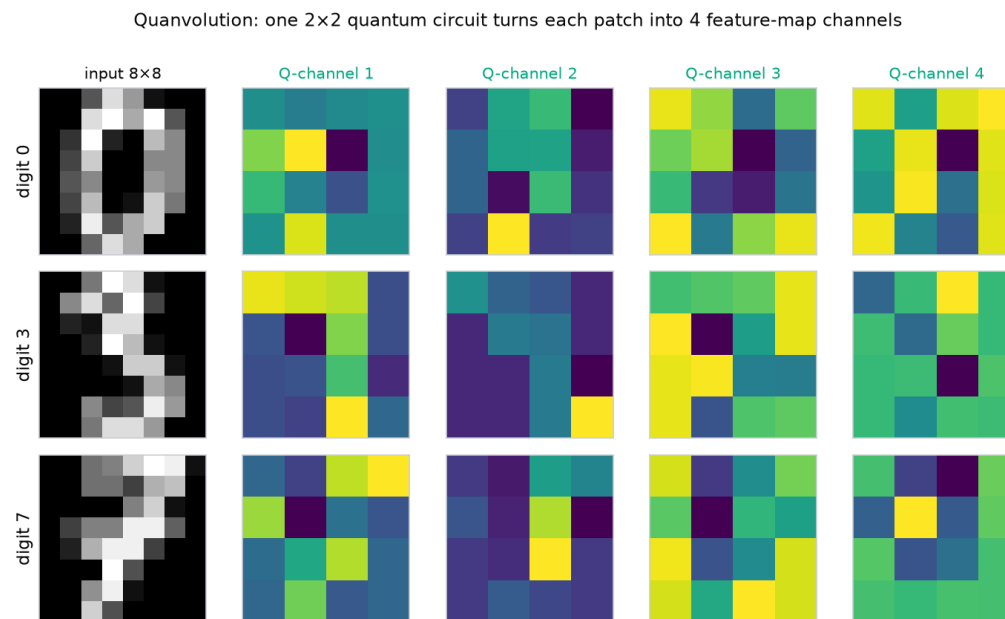
A hybrid: quantum front-end, classical back-end. The pattern for all near-term quantum vision.

THE ACTUAL CIRCUIT (PENNYLANE)



LIVE DEMO · QUANTUM FEATURE MAPS

One 2x2 circuit → four views of every digit.



Each measured channel is a different learned-by-physics view of the digit — the direct analogue of a classical CNN's feature maps, except **no training produced them; the entangling circuit did**. An untrained 4-qubit circuit preserved **~86%** of the classification signal.

The QCNN — a CNN built from qubits.

Quantum convolution

The same two-qubit unitary applied across neighbouring qubits — **weight sharing**, natively.

Quantum pooling

Measure some qubits, condition rotations on the outcome → **dimensionality reduction**.

The payoff

Only **$O(\log n)$** parameters, trainable, and relatively **resistant to barren plateaus**.

(Cong, Choi & Lukin, 2019.) Native use case: classifying **quantum states / phases of matter** — where the "image" is quantum data and there is **no classical shortcut**. For classical images it still needs encoding first.

FROM MY OWN WORK · CLASSICAL CNN → QUANTUM VISION

I built the CNN — then measured the NISQ wall.

J. PHYS.: CONF. SER. 2161 (2022) 012038

- Real CV pipeline: face detection → eye segmentation → **5-layer CNN** (batch-norm, ReLU, dropout) for eye-contact detection
- Explored the quantum frontier: N classical bits encode in **$\log_2 N$ qubits** — the amplitude-encoding claim, first-hand
- Ran a **Quantum Fourier Transform on real IBM hardware** — the core step of Quantum Visual Tracking

THE WALL, MEASURED

0.688

Input $|5\rangle \rightarrow$ measured **010** with probability 0.688, matching theory — with error **growing as circuit depth grows**. A first-hand NISQ-noise result.

Quantum Visual Tracking is the natural step past quanvolution — and where my research points.

THE HONEST PICTURE

No advantage on natural images yet — and why it still matters.

THE CAVEATS — SAY THEM OUT LOUD

- **On MNIST/CIFAR, tuned classical CNNs still win.** Quanvolution's gains are modest and setting-dependent.
- **Encoding cost can eat the speedup** — a megapixel image hits the same QRAM wall.
- **NISQ + shot noise** — expectation values need many measurements; demos run in simulation or on tiny patches.

WHERE THE PROMISE IS GENUINE

- **Quantum-native data** — QCNN classifies phases with no classical encoding
- structured / scientific imaging where a quantum kernel is hard to simulate
- **feature richness from $O(\log n)$ parameters** — for data-scarce scientific domains

Teach it as the front-end feature extractor of the future — running hybrid with the vision you already know.

Quantum is a pillar of Viksit Bharat 2047.

A developed India by the centenary of independence — and the frontier technologies that get us there.

THE ARGUMENT

- India already took frontier tech to **population scale** — Aadhaar, UPI, Digital India. That capability is proven.
- The next leap: **"world's back office"** → **"world's innovation engine."** Frugal jugaad is necessary but no longer sufficient.
- A coordinated mission stack: **NQM + IndiaAI + Supercomputing + Semicon + ANRF's ₹1 lakh crore RDI fund.**
- India's edge is the **algorithm & application layer** — exactly where QML, clustering and quantum vision live.

"Becoming a leader in quantum will prove central to Viksit Bharat — India can lead the era of quantum algorithm & application discovery."

— IBM, on India's path to Viksit Bharat 2047

2-3%
target R&D / GDP by
2047

2047
centenary of
independence

THE CLOSE · YOUR MOVE

The bottleneck is people. You are the lever.

Teach it

Convolution to quanvolution is a first-principles story — the demo runs in simulation on a laptop. Seed one module, one project, one reading group. The global shortage of people who understand this stack is India's opening.

Build with it

The notebooks from today run on a laptop — PennyLane, scikit-learn, no hardware needed. Point a student at them this week.

Connect to the mission

NQM hubs, state quantum valleys and startups (QpiAI, QNu Labs) need talent. Your classroom is the top of that funnel.

Viksit Bharat 2047 will be built by the cohort sitting in your classrooms right now. Every course you seed is India's comparative advantage compounding.

thank you

Archit Srivastava · QML-2026 FDP · NIT Warangal × NIT Raipur

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